

# Statistical and Machine Learning

Evaluation of model performance

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# Evaluation of model performance

- ▶ Evaluation
  - ▶ Evaluation metric
  - ▶ Cross validation
  - ▶ Bias-variance trade off
  - ▶ Bootstrap

# Importance of evaluation

- ▶ Evaluation metric is closely associated with the goal of model
- ▶ Example:
  - ▶ For classification problems, we often consider high accuracy as our goal
  - ▶ Consider we build two models to classify cancer and health patient based on some clinical variables. Both have accuracy of 110/140. Which is better?
    - ▶ **I need more information to answer this question!**

# Importance of evaluation

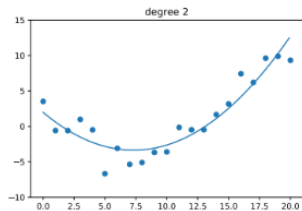
- ▶ What is the cost of predicting a healthy person to be a patient with cancer?
  - ▶ **Not that big** – apart from financial burden a healthy person will have some extra checks done.
- ▶ What is the cost of predicting a person with cancer to be healthy?
  - ▶ **Enormous cost** – a patient with cancer might suffer fatal consequences due to the wrong prediction.
- ▶ Cancer samples are rare, but we have the same weight for healthy people and patients with cancer.

| Prediction True | Health | Cancer |
|-----------------|--------|--------|
| Health          | 100    | 20     |
| Cancer          | 10     | 10     |

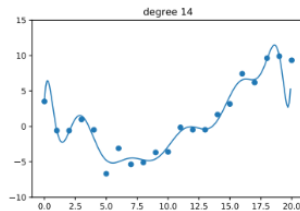
| Prediction True | Health | Cancer |
|-----------------|--------|--------|
| Health          | 90     | 30     |
| Cancer          | 0      | 20     |

# Why evaluation is important?

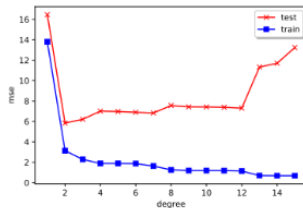
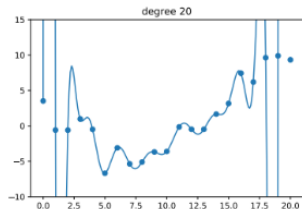
- ▶ Risk of overfitting – one cannot generalise beyond data at hand
- ▶ Example: true model  $y = -\frac{3}{2}x + \frac{1}{9}x^2 + \epsilon$  (top left)



(a)



(b)



# Evaluation of model performance

- ▶ Evaluation
  - ▶ **Evaluation metric**
  - ▶ Cross validation
  - ▶ Bias-variance trade off
  - ▶ Bootstrap

# Evaluation metric

- ▶ Classification
  - ▶ Confusion matrix
  - ▶ TPR, FPR, Sensitivity, Specificity, Precision, Recall and F1 score
  - ▶ ROC curve and PR curve
- ▶ Regression
  - ▶  $l_p$  loss
  - ▶ huber loss

# Evaluation metric

- ▶ Classification
  - ▶ **Confusion matrix**
  - ▶ **TPR, FPR, Sensitivity, Specificity, Precision, Recall and F1 score**
  - ▶ ROC curve and AUC
- ▶ Regression
  - ▶ lp loss
  - ▶ huber loss

# Confusion matrix

- ▶ TP: True positive (**correct**) vs FP: False positive (**incorrect**)
- ▶ TN: True negative (**correct**) vs FN: False negative (**incorrect**)

|                  |              | Actual Values |              |
|------------------|--------------|---------------|--------------|
|                  |              | Positive (1)  | Negative (0) |
| Predicted Values | Positive (1) | TP            | FP           |
|                  | Negative (0) | FN            | TN           |

# Indicators of the goodness of fit

|                  |              | Actual Values |              |
|------------------|--------------|---------------|--------------|
|                  |              | Positive (1)  | Negative (0) |
| Predicted Values | Positive (1) | TP            | FP           |
|                  | Negative (0) | FN            | TN           |

## Accuracy:

$$acc = \frac{TP+TN}{TP+TN+FP+FN}$$

## True positive rate/Sensitivity:

$$TPR = R = Sensitivity = \frac{TP}{TP+FN}$$

## False positive rate/Type I error:

$$FPR = \frac{FP}{FP+TN}$$

## Specificity:

$$Specificity = 1 - FPR = \frac{TN}{FP+TN}$$

## Precision:

$$P = \frac{TP}{TP+FP}$$

## F1-Score:

$$F1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2 \times P \times R}{P+R}$$

# Evaluation metric

- ▶ Classification
  - ▶ Confusion matrix
  - ▶ TPR, FPR, Sensitivity, Specificity, Precision, Recall and F1 score
  - ▶ **ROC curve and AUC**
- ▶ Regression
  - ▶  $l_p$  loss
  - ▶ huber loss

# ROC Curve and AUC

- ▶ A classifier often outputs **scores**  $\hat{y}_i \in [0, 1]$
- ▶ To obtain class predictions, we choose a threshold  $t$ :
  - ▶ If  $\hat{y}_i \geq t \rightarrow$  predict positive
  - ▶ If  $\hat{y}_i < t \rightarrow$  predict negative
- ▶ Different thresholds produce different values of TPR and FPR
- ▶ Instead of choosing one threshold, evaluate the model over many thresholds
- ▶ ROC = Receiver Operating Characteristic

**ROC curve:** plot  $(\text{FPR}(t), \text{TPR}(t))$  for thresholds  $t \in [0, 1]$

## Area Under the Curve (AUC)

AUC provides a **single-number summary** and measures how well the model **ranks positives above negatives**.

Interpretation:

- ▶ 1 = perfect classifier
- ▶ 0.5 = random guessing

## ROC and AUC – examples

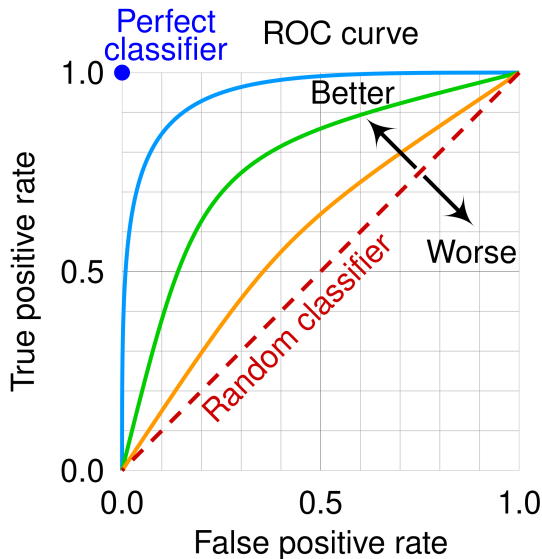


Image taken from [Wikipedia](#).

# Evaluation metric

- ▶ Classification
  - ▶ Confusion matrix
  - ▶ TPR, FPR, Sensitivity, Specificity, Precision, Recall and F1 score
  - ▶ ROC curve and AUC
- ▶ Regression
  - ▶ **lp loss**
  - ▶ huber loss

## Lp loss

- ▶ For regression, we also consider different types of error. A general formulation of the loss function is the so-called  $L_p$  loss:

$$L_p = \sum_{i=1}^n |y_i - \hat{y}_i|^p$$

- ▶ If  $p = 1$ , we obtain the  $L_1$  loss (absolute value loss). It is robust to outliers but non-differentiable.

$$L_1 = \sum_{i=1}^n |y_i - \hat{y}_i|$$

- ▶ If  $p = 2$ , we obtain the  $L_2$  loss (least squares loss), i.e., one of the most widely used loss functions due to its desirable properties.

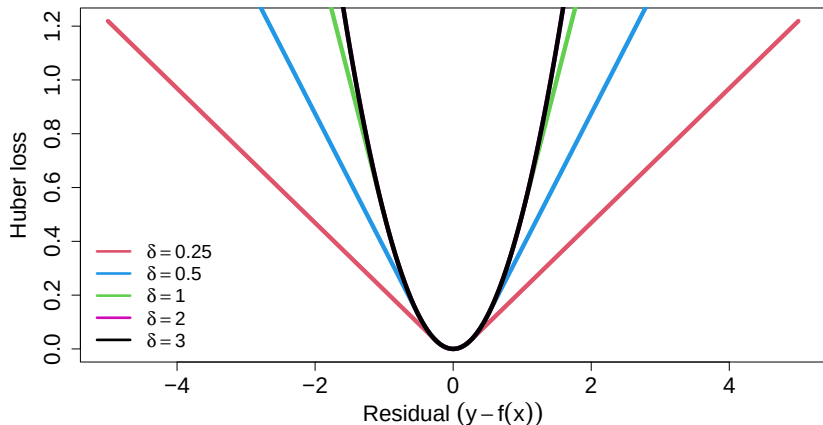
$$L_2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

## Huber loss

Combine L1 and L2 loss  $\rightarrow$  Huber loss

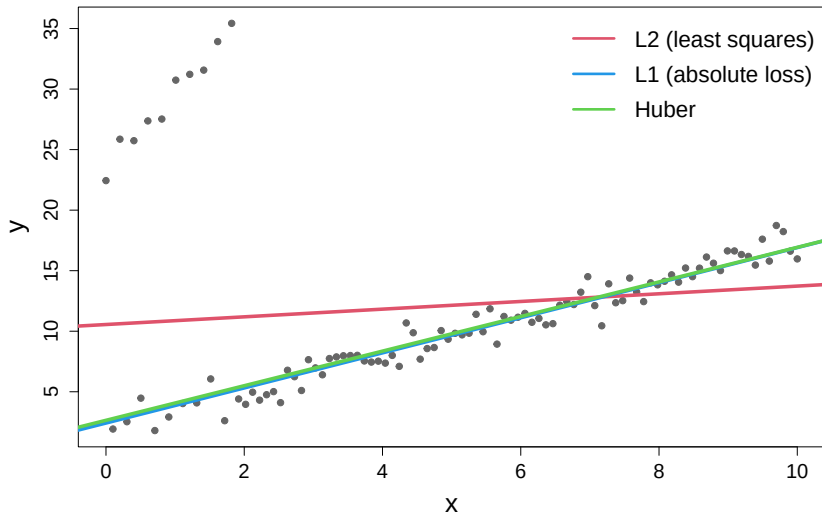
$$L_{\delta}(y, f(x)) = \begin{cases} \frac{1}{2}(y - f(x))^2, & |y - f(x)| \leq \delta \\ \delta \left( |y - f(x)| - \frac{1}{2}\delta \right), & \text{otherwise} \end{cases}$$

Huber loss for different  $\delta$



# Effect of influential observations /on regression fits

## Linear fits



# Evaluation of model performance

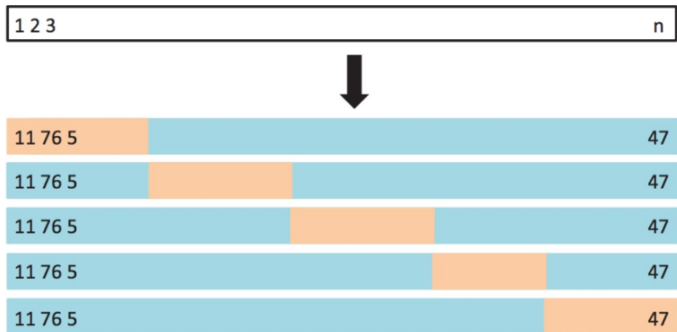
- ▶ Evaluation
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  - ▶ **Cross validation**
  - ▶ Bias-variance trade off
  - ▶ Bootstrap

# Cross validation

- ▶ Idea: randomly divide the data into  $K$  equal-sized parts
- ▶ We leave out part  $k$ , fit the model to the other  $K - 1$  parts (combined), and then obtain predictions for the left-out  $k$ th part.
- ▶ This is done separately for each part  $k = 1, 2, \dots, K$  and then the results are combined.

# Cross validation

## ► Illustration: 5-fold Cross validation



## Example: estimating the mean

Suppose we observe data

$$X_1, X_2, \dots, X_n$$

and want to estimate the population mean

$$\mu = \mathbb{E}[X].$$

5-fold cross-validation procedure:

1. Split the data into 5 folds:

$$\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5$$

2. For each fold  $k = 1, \dots, 5$ , compute the mean using the data **excluding** fold  $k$ :

$$\hat{\mu}^{(-k)} = \frac{1}{n - n_k} \sum_{i \notin \mathcal{D}_k} X_i$$

# Cross-Validation Estimator

Cross-validated estimator of the mean:

$$\hat{\mu}_{CV} = \frac{1}{5} \sum_{k=1}^5 \hat{\mu}^{(-k)}$$

For each fold, compute the validation error:

$$CV_k = \frac{1}{n_k} \sum_{i \in \mathcal{D}_k} \left( X_i - \hat{\mu}^{(-k)} \right)^2$$

Cross-validation estimate of prediction error:

$$CV_{(5)} = \frac{1}{5} \sum_{k=1}^5 CV_k$$

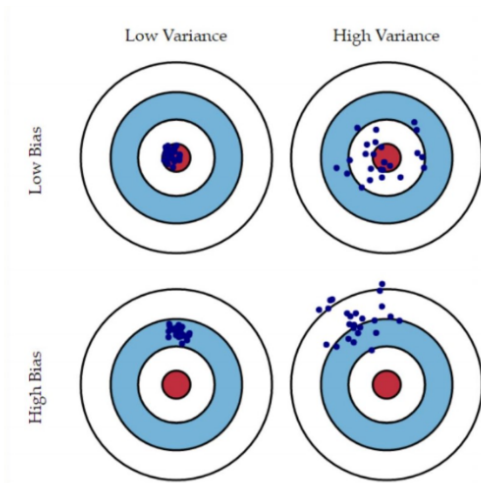
# Evaluation

- ▶ Evaluation
  - ▶ Evaluation metric
  - ▶ Cross validation
  - ▶ **Bias-variance trade off**
  - ▶ Bootstrap

# Bias-Variance trade off

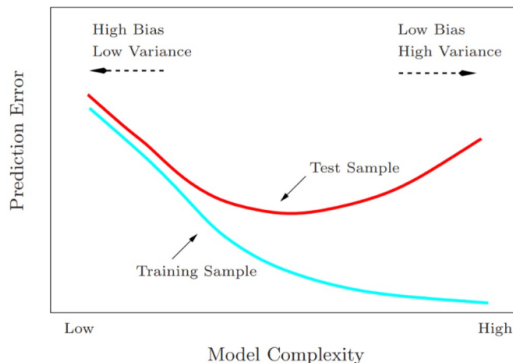
**Mean squared error (MSE)**

$$MSE = Bias^2 + Var$$



# Bias-Variance trade off

- ▶ As model complexity increases, training error decreases; bias decreases but variance increases.
- ▶ Test error typically decreases first and then increases (U-shape): simple models underfit (high bias), complex models overfit (high variance).

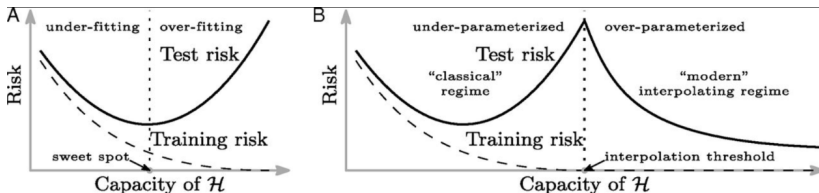


# Modern ML/AI algorithms – double descent

## Double descent phenomenon

As model complexity increases:

1. Test error decreases (model learns signal)
2. Test error increases near the interpolation threshold
3. Test error decreases again for highly overparameterized models



The peak occurs when the model just becomes flexible enough to perfectly fit the training data.

# Why Does the Second Descent Appear?

After the interpolation threshold:

- ▶ There are many models that perfectly fit the training data.
- ▶ Optimization methods tend to select smooth or minimum-norm solutions (implicit regularization).
- ▶ These solutions often have lower variance and generalize better.
- ▶ As model complexity increases further, effective complexity may decrease, and test error decreases again.

Key idea: Model size and effective model complexity are not the same.

## Summary: Double Descent Across Methods

Double descent mainly appears in models that can interpolate the data and where optimization selects smooth or minimum-norm solutions.

| Method                        | DD?   | Why                              |
|-------------------------------|-------|----------------------------------|
| Linear regression ( $p > n$ ) | Yes   | Min-norm interpolating solution  |
| Polynomial regression         | Yes   | High-degree models interpolate   |
| Neural networks               | Often | Implicit regularization          |
| Decision trees                | Rare  | No smooth interpolating solution |

### Key takeaway:

Double descent is mainly associated with interpolating, overparameterized models and implicit regularization.

# Evaluation

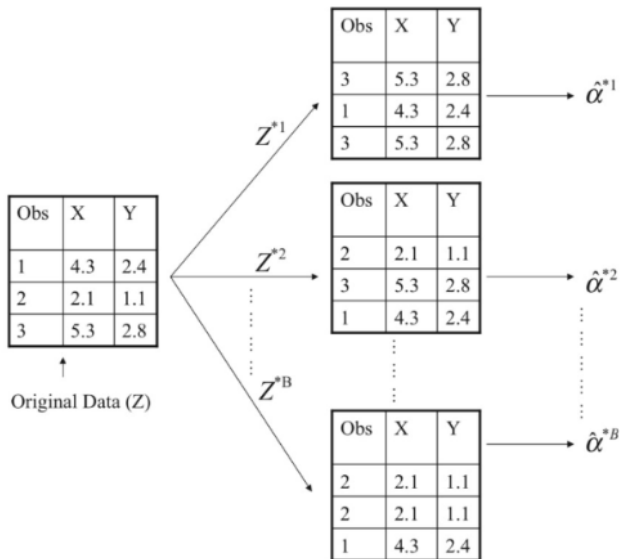
- ▶ Evaluation
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  - ▶ Bias-variance trade off
  - ▶ **Bootstrap**

# Bootstrap

- ▶ How to quantify variance/uncertainty? → Bootstrap!
- ▶ The bootstrap is a flexible and powerful statistical tool that can be used to quantify the uncertainty associated with a given estimator or statistical learning method.
- ▶ For example, if done correctly and under certain assumptions, it can provide an estimate of the standard error of a target parameter, or a confidence interval for that target parameter.

# Bootstrap

- Essentially sampling with replacement



## Simple example

- ▶ Suppose that we wish to invest a fixed sum of money in two financial assets that yield returns of  $X$  and  $Y$  where  $X$  and  $Y$  are random quantities.
- ▶ The goal is to create a portfolio by investing fraction  $\alpha$  of our wealth in  $X$  and  $(1 - \alpha)$  in  $Y$ .
- ▶ We want to minimise the total risk of the investment. Mathematically, this involves minimizing  $\text{Var}(\alpha X + (1 - \alpha)Y)$
- ▶ The solution to this problem (calculus) is:

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

$$\sigma_X^2 = \text{Var}(X), \sigma_Y^2 = \text{Var}(Y); \sigma_{XY} = \text{Cov}(X, Y)$$

## Simple example

- ▶ The values of  $\sigma_X^2$ ,  $\sigma_Y^2$  and  $\sigma_{XY}$  are unknown but estimates can be computed from the data.
- ▶ The estimate of  $\alpha$  that minimizes the variance of the investment can then be computed with:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$

- ▶ Suppose that  $X$  and  $Y$  can be sampled from the population repeatedly.
- ▶ To estimate the standard deviation of  $\hat{\alpha}$ , paired observations  $(X, Y)$  can be repeatedly sampled, say 100 pairs to get a single estimate of  $\alpha$ . Repeat this process to get 1000 estimates for  $\alpha$ .
- ▶ Denote these estimates  $\hat{\alpha}_1, \dots, \hat{\alpha}_{1000}$
- ▶ **Key ideas:**
- ▶ reproduce faithfully the population you are sampling from.
- ▶ sample in a “suitable” way

# Bootstrap simulations

- ▶ Consider example with  $\sigma_X^2 = 1$ ,  $\sigma_Y^2 = 1.5$  and  $\sigma_{XY} = 0.5 \rightarrow \alpha = \frac{2}{3}$

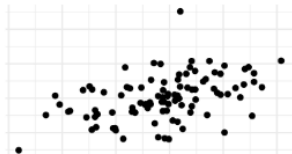
Simulation: 1



Simulation: 2



Simulation: 3



Simulation: 4



- ▶ Each panel shows 100 simulated returns. From left to right, top to bottom, the estimates for  $\alpha$  are 0.659, 0.683, 0.726, 0.68.

## Parameter estimates

- ▶ Consider the mean of all the parameter estimates

$$\bar{\hat{\alpha}} = \frac{1}{1000} \sum_{k=1}^{1000} \hat{\alpha}_k = 0.6662595$$

- ▶ This is close to the true value of 0.6666667
- ▶ Estimate of the standard error using the standard deviation of all the estimates:

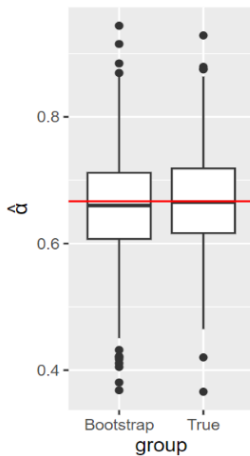
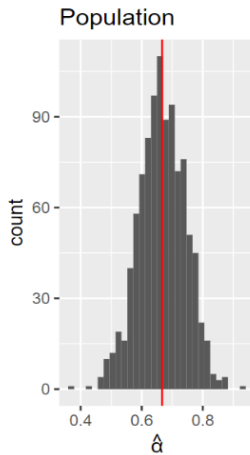
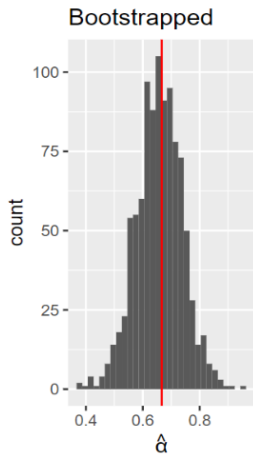
$$\sqrt{\frac{1}{1000-1} \sum_{k=1}^{1000} (\hat{\alpha}_k - \bar{\hat{\alpha}})^2} = 0.0760217$$

- ▶ This gives an intuitive description of the reliability of the estimator.
- ▶ For a random sample the estimate would vary around the true value by 0.0760217.
- ▶ Under certain assumptions, one can prove that these estimates converge to true values.

## Application in reality

- ▶ We cannot apply this directly in reality, as we do not know the population we are sampling from.
- ▶ Bootstrap attempts to mimic this process within two essential steps: reproducing the population and sampling in a suitable way.
- ▶ In case of i.i.d data, instead of sampling new independent observations from the population, resample observations from the data with replacement.
- ▶ Some observations appear more than once and some not at all.

# Results bootstrap vs population



# Summary

- ▶ Choose a suitable evaluation metric
- ▶ Cross validation to evaluate different models
- ▶ Bootstrap to measure uncertainty of parameter estimation

# References

- ▶ [ESLII](#): Chapters 7, 8.2
- ▶ [PRML](#) Chapters 3.2
- ▶ [PML](#): Chapters 4.5.5, 4.7.3, 5.1.3